

THE UNIVERSITY OF LAHORE

Project Proposal

Blood Donation Management System

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Course:	Data Structures and Algorithms (DSA)
Technology:	TypeScript, Node.js, MongoDB

1. Introduction

The Blood Donation Management System is a simple and user-friendly system designed to help patients find blood donors quickly during emergencies. The system stores donor information such as name, blood group, city, and contact number, allowing users to search for suitable donors efficiently. This project focuses on implementing Data Structures and Algorithms concepts for managing donor records in an organized and efficient manner.

2. Problem Statement

Finding blood donors during emergencies is often difficult and time-consuming. Patients and hospitals usually depend on social media or personal contacts to search for blood donors. This process may delay treatment and create serious problems. Therefore, a digital system is required to manage donor records and blood requests efficiently.

3. Objectives

- To create a digital blood donor management system.
- To store donor information efficiently.
- To search donors by blood group and city.

- To manage blood requests easily.
- To apply DSA concepts in a real-world project.

4. Scope of the Project

The project will allow users to register as donors, search available donors, and submit blood requests. The system will also help manage donor records digitally. This project can later be expanded with advanced features such as notifications and location tracking.

5. DSA Concepts Used

- Arrays and Lists for storing records.
- Searching Algorithms for finding donors.
- Sorting donor records alphabetically.
- CRUD Operations for managing data.

6. Functional Requirements

- Add donor information.
- Display donor records.
- Search donors by blood group.
- Add blood requests.
- Delete donor records.

7. Technologies Used

Frontend: Ionic Angular, HTML, CSS, TypeScript

Backend: Node.js and Express.js

Database: MongoDB

8. Expected Outcomes

The system will help patients find blood donors quickly and manage donor information digitally. It will reduce the time required to search for blood donors and improve emergency response efficiency.

9. Conclusion

The Blood Donation Management System is a useful and simple project that combines practical problem solving with Data Structures and Algorithms concepts. It provides an efficient way to manage donor records and blood requests digitally.